

Final Exam – Medicinal Plants

Lectures 5–12 (Academic Model)

Time: 3 Hours

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Total Marks: 100

Marking Scheme:

- Each question = 2 marks
- Total questions = 50

Instructions: - Answer all questions. - Choose the most appropriate answer. - Questions cover Lectures 5 to 12 only. - Difficulty and question types are mixed.



Section A: MCQs – Mixed Concepts (40 Questions)

Q1.

The leaf is best described as:

- A. A reproductive organ
- B. A lateral outgrowth from the stem
- C. A modified root
- D. An underground stem

Q2.

Paracytic stomata are characterized by:

- A. No subsidiary cells
- B. Subsidiary cells parallel to guard cells
- C. Subsidiary cells perpendicular to guard cells
- D. Unequal subsidiary cells

Q3.

The laxative action of Senna is mainly due to:

- A. Direct irritation of stomach
- B. Anthraquinone glycosides
- C. Flavonoids
- D. Alkaloids

Q4.

Sennosides are considered prodrugs because they:

- A. Are active in stomach
- B. Act immediately after ingestion
- C. Are activated by colonic bacteria
- D. Are excreted unchanged

Q5.

The anticholinergic action of Belladonna is due to:

- A. Increase acetylcholine release
- B. Block muscarinic receptors
- C. Stimulate parasympathetic system
- D. Increase gastric secretion

Q6.

Which plant shows strong anticholinergic toxicity?

- A. Senna
- B. Datura
- C. Uva Ursi
- D. Digitalis

Q7.

Hyoscyamus alkaloids mainly act by:

- A. Stimulating CNS
- B. Blocking acetylcholine
- C. Increasing dopamine
- D. Blocking adrenergic receptors

Q8.

Vitali–Morin reaction is specific for:

- A. Cardiac glycosides
- B. Tropane alkaloids
- C. Anthraquinones
- D. Saponins

Q9.

Uva Ursi is mainly used in:

- A. Cardiac failure
- B. Bronchitis
- C. Urinary tract infections
- D. Constipation

Q10.

The antibacterial action of Uva Ursi is mainly due to:

- A. Tannins
- B. Hydroquinone
- C. Flavonoids
- D. Saponins

Q11.

Guava leaves are rich in:

- A. Alkaloids
- B. Cardiac glycosides
- C. Tannins
- D. Anthraquinones

Q12.

Eucalyptus oil is mainly used as:

- A. Laxative
- B. Cardiotonic
- C. Expectorant
- D. Sedative

Q13.

Which plant contains saponins useful in productive cough?

- A. Guava
- B. Ivy
- C. Senna
- D. Tea

Q14.

The official *Digitalis* species is:

- A. *Digitalis lanata*
- B. *Digitalis purpurea*
- C. *Digitalis lutea*
- D. *Digitalis alba*

Q15.

Digitalis glycosides exert their effect by:

- A. Reducing heart rate
- B. Increasing myocardial contractility
- C. Vasodilation
- D. Diuresis only

Q16.

Lanatosides differ from purpurea glycosides by:

- A. Hydroxyl group
- B. Sulfate group
- C. Acetyl group
- D. Methyl group

Q17.

Tea leaves are obtained from:

- A. Roots
- B. Mature lower leaves
- C. Young leaves and buds
- D. Stem bark

Q18.

The stimulant effect of tea is due to:

- A. Tannins
- B. Flavonoids
- C. Caffeine
- D. Saponins

Q19.

Fungi differ from plants mainly by absence of:

A. Cell wall B. Nucleus C. Chlorophyll D. Cytoplasm

Q20.

The cell wall of fungi contains:

A. Cellulose B. Lignin C. Chitin D. Pectin

Section B: Reasoning & Tricky MCQs (20 Questions)

Q21.

Senna is contraindicated in intestinal obstruction because it:

A. Decreases water content B. Increases peristalsis and water
C. Causes sedation D. Reduces enzyme secretion

Q22.

Anticholinergic crisis is characterized by all EXCEPT:

A. Dry skin B. Dilated pupils C. Excess salivation D. Hyperthermia

Q23.

Hyoscine is preferred clinically because it:

A. Has strong cardiac action B. Produces CNS sedation
C. Is free from side effects D. Is rapidly excreted

Q24.

Uva Ursi should not be used long-term mainly due to:

A. Loss of efficacy B. High tannin content C. Hypokalemia D. Cardiac toxicity

Q25.

Digitalis requires careful dose monitoring because it:

A. Is water soluble B. Has narrow therapeutic index
C. Is rapidly metabolized D. Is weakly active

Section C: Case-Based MCQs (20 Questions)

Q26.

A patient with motion sickness is best treated with:

A. Atropine B. Hyoscine C. Senna D. Ivy

Q27.

A powdered leaf drug showing anomocytic stomata and beaded anticlinal walls suggests:

A. Senna B. Digitalis lanata C. Guava D. Uva Ursi

Q28.

Greenish-brown discoloration of urine during treatment indicates use of:

A. Senna B. Digitalis C. Uva Ursi D. Ivy

Q29.

A patient with heart failure should avoid overdose of:

A. Senna B. Belladonna C. Digitalis D. Guava

Q30.

Excessive dryness of mouth and blurred vision suggest toxicity of:

A. Uva Ursi B. Datura C. Senna D. Tea

Section D: Taxonomy & Classification (20 Questions)

Q31.

Plant taxonomy deals mainly with:

A. Physiology B. Identification and classification C. Pharmacology D. Cultivation

Q32.

Cryptogams reproduce mainly by:

A. Seeds B. Flowers C. Spores D. Fruits

Q33.

Gymnosperms are characterized by:

A. Covered seeds B. Fruits C. Naked seeds D. Spores only

Q34.

Monocotyledons have:

A. Two cotyledons B. Parallel venation C. Reticulate venation D. Secondary growth

Q35.

The lowest taxonomic rank is:

A. Family B. Genus C. Species D. Order

Answer Key (End of File)

1-B | 2-B | 3-B | 4-C | 5-B | 6-B | 7-B | 8-B | 9-C | 10-B

11-C | 12-C | 13-B | 14-B | 15-B | 16-C | 17-C | 18-C | 19-C | 20-C

21-B | 22-C | 23-B | 24-B | 25-B

26-B | 27-B | 28-C | 29-C | 30-B

31-B | 32-C | 33-C | 34-B | 35-C

End of Final Exam